Phase 8: Project Demonstration

SmartLender: AI-Powered Credit Card Approval System

Demo Scenario 1 – Automated Screening

A credit analyst enters an applicant's financial profile into the web application. The model returns an instant approve/reject prediction, letting the analyst prioritize which applications need closer review.

Demo Scenario 2 – High-Risk Identification

A compliance officer screens applicants with past-due loan records. The feature engineering pipeline converts multi-class payment status into a binary risk label, so high-risk applicants are clearly flagged.

Demo Scenario 3 – Self-Service Eligibility Check

A prospective customer enters their own details before formally applying and receives an instant likelihood of approval, reducing the number of applications that would otherwise be rejected outright.

Conclusion & Future Scope

SmartLender shows that a tuned XGBoost model can automate a meaningful share of the credit approval workload. Future work includes adding explainability for each decision and monitoring the model for drift as new applicant data comes in.

Terminal Execution

The screenshot below shows the Flask development server starting successfully and serving requests on local port 5000.

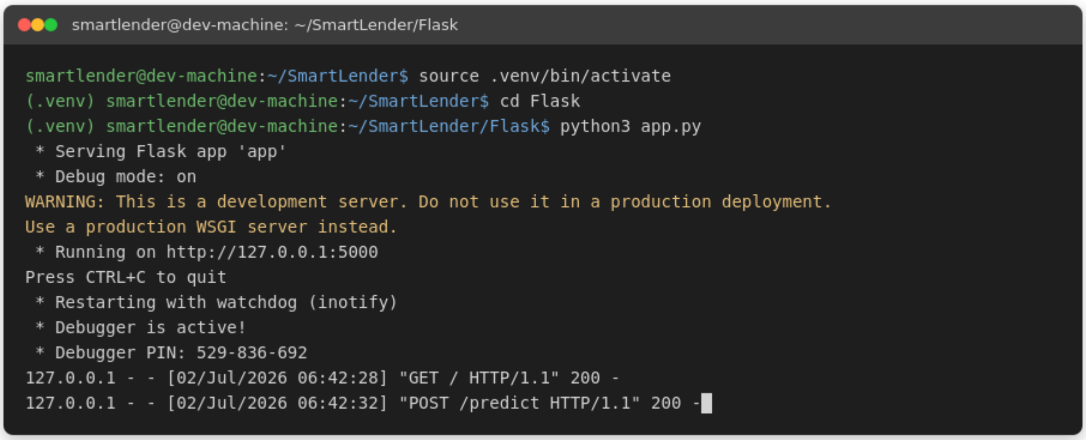

Figure 5: Flask server running in terminal

Web Interface – Input Form

The applicant enters demographic and financial details through the web form shown below.

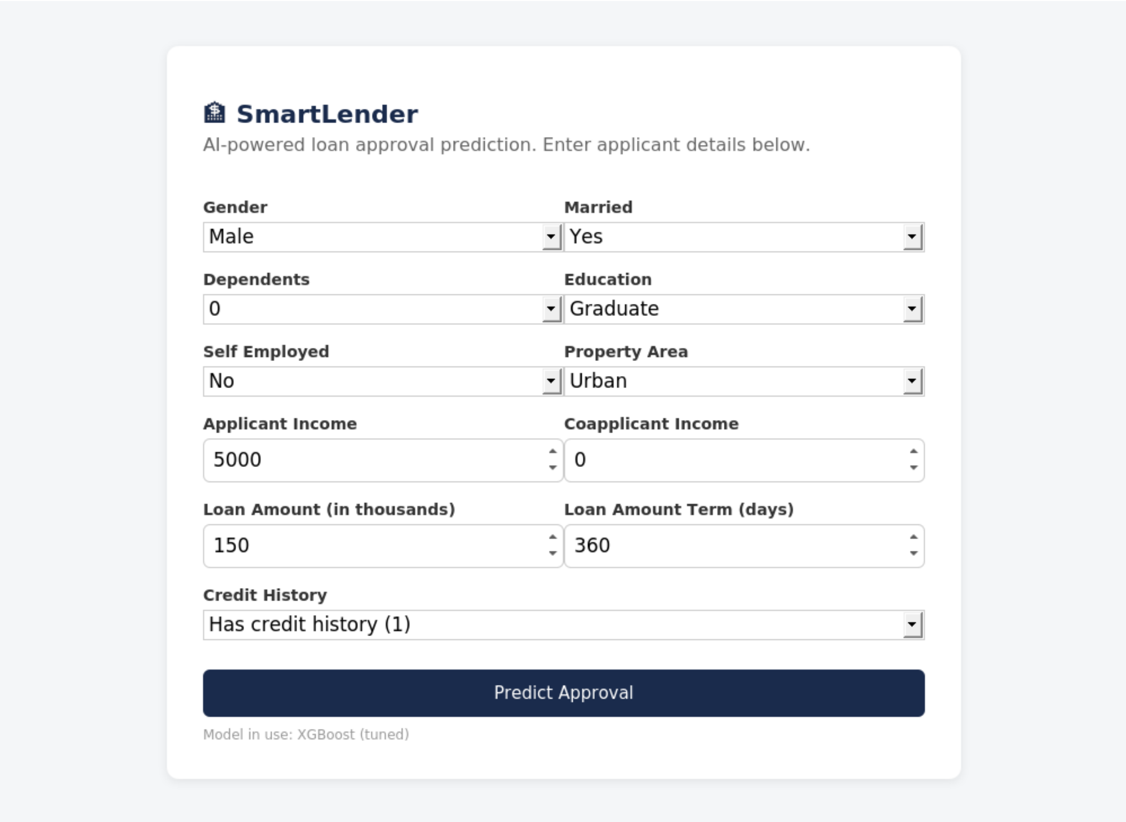

Figure 6: Input form interface

Web Interface – Prediction Output

After submission, the system returns the approve/reject decision along with the tuned XGBoost model's confidence percentage.

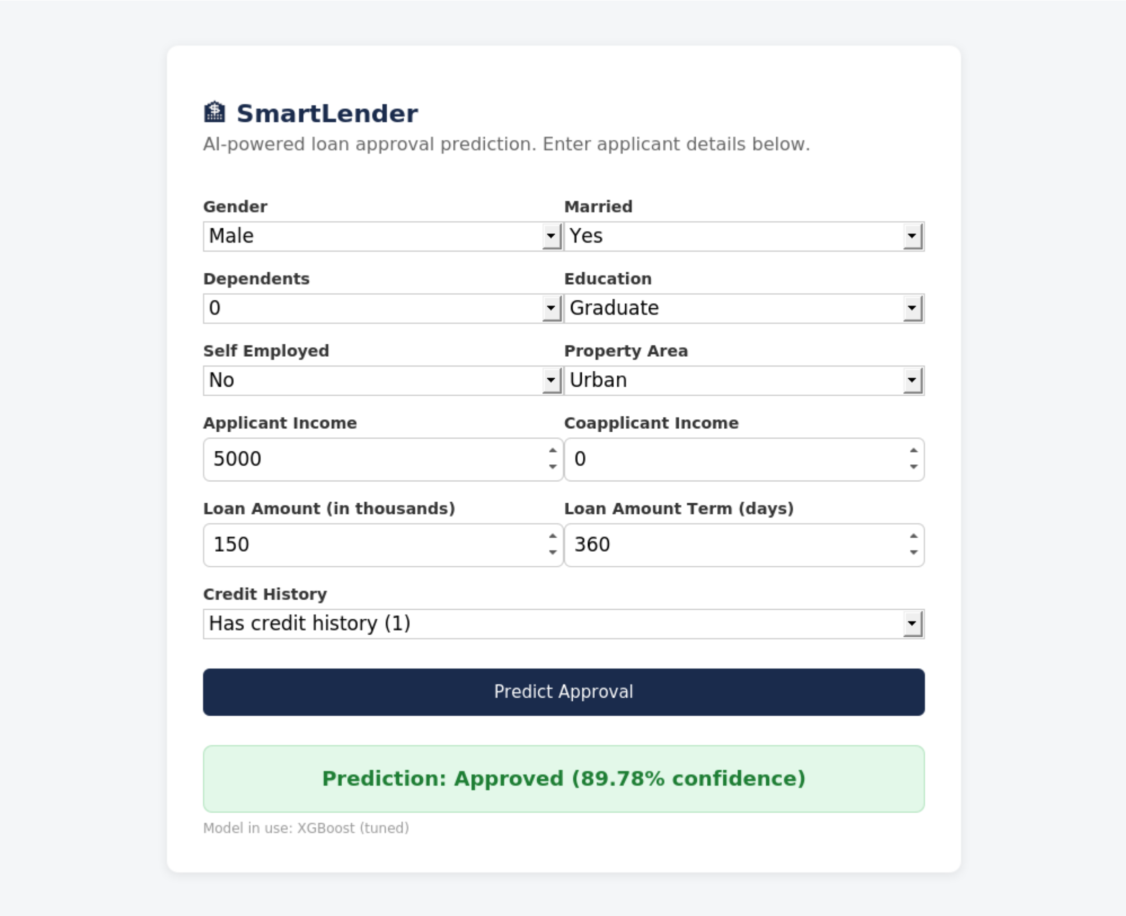

Figure 7: Prediction output display